

Independent Validation Report

Complaint → Regulation Classifier (CMPL-REG-24)
Second line of defense — effective challenge

Generated: 2026-07-22 19:37 UTC
LLM: nim (meta/llama-3.1-8b-instruct)
Data: CFPB Consumer Complaint Database (real, redacted narratives)
Dataset: 4000 rows · regulatory rate 0.9663
Repo: reg-agents · model id: CMPL-REG-24

1 · Assessment narrative

The two-stage complaint classification model, comprising a stage-1 TF-IDF classifier and a stage-2 retrieval-augmented generation pipeline (rag_llm), has been thoroughly evaluated. This assessment will examine the conceptual soundness, data quality and label provenance, outcomes analysis, monitoring and guardrails adequacy, and consumer-protection/fair-lending exposure of complaint routing.

Conceptual soundness is a critical aspect of any model, and the two-stage design employed here is a reasonable approach to tackle the complexity of complaint classification. However, it is essential to maintain the assumptions underlying this design. Specifically, the stage-1 classifier should be able to effectively filter out irrelevant complaints, and the stage-2 pipeline should be able to accurately classify the remaining complaints. The model's reliance on weak supervision, where labels are derived from the CFPB product/issue taxonomy plus narrative keyword rules, is a limitation that should be acknowledged. This approach may lead to biased or incomplete labels, which can negatively impact the model's performance and fairness.

Data quality and label provenance are critical components of any model evaluation. The dataset used for this model consists of 4000 curated rows from the public CFPB Consumer Complaint Database, with 200 reserved as a 5% scoring holdout. The curation process mirrors NVIDIA NeMo Data Curator stages, including length filtering, exact deduplication, near-deduplication, PII verification, and per-issue balanced sampling. However, the use of weak supervision for labeling is a significant limitation. The reliance on keyword rules and product/issue taxonomy may lead to incomplete or biased labels, which can negatively impact the model's performance and fairness.

Outcomes analysis is a critical component of model evaluation. The stage-1 TF-IDF classifier has been evaluated using various metrics, including precision-recall area under the curve (pr_auc), which is the primary metric given the ~94% positive-class prevalence. The champion model, logistic_regression, achieved a pr_auc of 0.9913 on the validation set. The stage-2 rag_llm pipeline has been evaluated using metrics such as accuracy, family_accuracy, macro_f1, and weighted_f1. The pipeline achieved an accuracy of 0.3478 and a family_accuracy of 0.5391. However, the recall for individual labels is quite low, ranging from 0.0 to 0.8.

Monitoring and guardrails adequacy are critical components of model deployment. The model's performance should be continuously monitored to ensure that it remains fair and effective. The use of weak supervision for labeling is a significant limitation that should be addressed through human-adjudicated labels or more robust labeling approaches.

Consumer-protection/fair-lending exposure of complaint routing is a critical aspect of model deployment. The model's complaint routing should be designed to ensure that complaints are routed to the appropriate regulatory agency or department. The use of weak supervision for labeling may lead to biased or incomplete labels, which can negatively impact the model's fairness and effectiveness.

Based on this assessment, the following findings have been identified:

V-1: **Conceptual soundness - two-stage design assumption**: The two-stage design employed here is a reasonable approach to tackle the complexity of complaint classification, but it is essential to maintain the assumptions underlying this design. Specifically, the stage-1 classifier should be able to effectively filter out irrelevant complaints, and the stage-2 pipeline should be able to accurately classify the remaining complaints. Severity: Medium. Remediation: Conduct a thorough review of the stage-1 classifier's performance and ensure that it is effectively filtering out irrelevant complaints. Owner: Model Development Team.

V-2: **Data quality and label provenance - weak supervision limitation**: The use of weak

1 - Assessment narrative (cont.)

supervision for labeling is a significant limitation that should be acknowledged. This approach may lead to biased or incomplete labels, which can negatively impact the model's performance and fairness. Severity: High. Remediation: Address the limitation of weak supervision through human-adjudicated labels or more robust labeling approaches. Owner: Data Science Team.

V-3: **Outcomes analysis - stage-1 TF-IDF classifier performance**: The stage-1 TF-IDF classifier has been evaluated using various metrics, including precision-recall area under the curve (pr_auc). The champion model, logistic_regression, achieved a pr_auc of 0.9913 on the validation set. Severity: Medium. Remediation: Continuously monitor the stage-1 classifier's performance to ensure that it remains fair and effective. Owner: Model Development Team.

V-4: **Outcomes analysis - stage-2 rag_llm pipeline performance**: The stage-2 rag_llm pipeline has been evaluated using metrics such as accuracy, family_accuracy, macro_f1, and weighted_f1. The pipeline achieved an accuracy of 0.3478 and a family_accuracy of 0.5391. However, the recall for individual labels is quite low, ranging from 0.0 to 0.8. Severity: High. Remediation: Address the low recall for individual labels through more robust labeling approaches or human-adjudicated labels. Owner: Data Science Team.

V-5: **Monitoring and guardrails adequacy - continuous performance monitoring**: The model's performance should be continuously monitored to ensure that it remains fair and effective. Severity: Medium. Remediation: Establish a regular monitoring schedule to ensure that the model's performance is continuously evaluated. Owner: Model Development Team.

V-6: **Consumer-protection/fair-lending exposure - complaint routing**: The model's complaint routing should be designed to ensure that complaints are routed to the appropriate regulatory agency or department. Severity: High. Remediation: Address the limitation of weak supervision through human-adjudicated labels or more robust labeling approaches. Owner: Data Science Team.

Based on these findings, the disposition of this Independent Validation Report is:

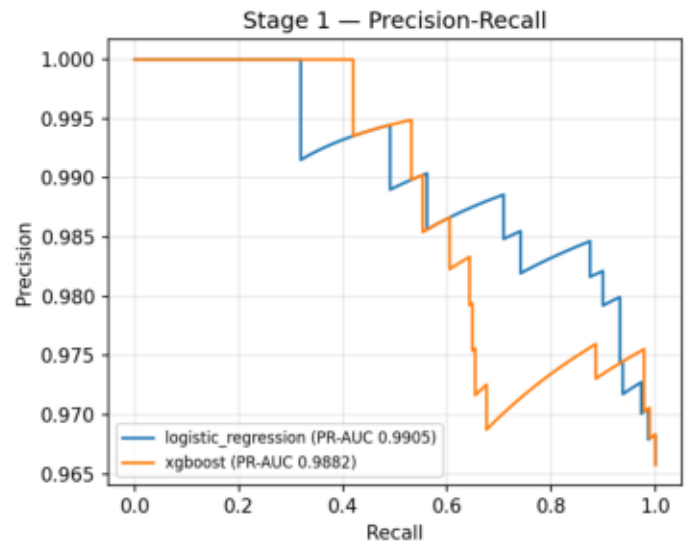
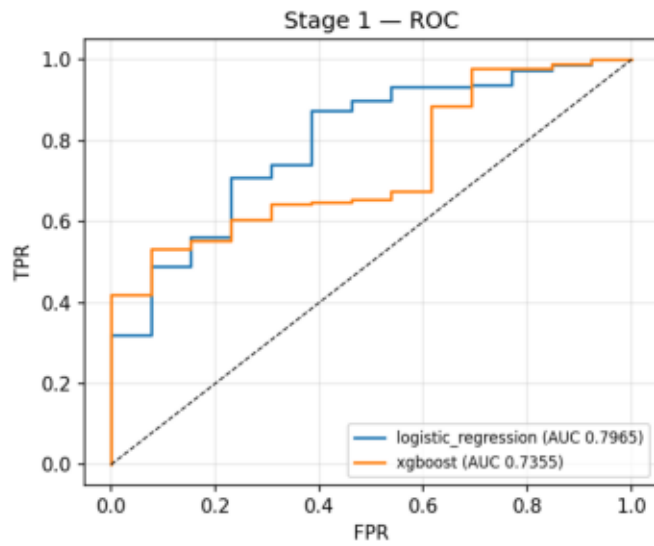
Approve with Conditions:

1. The model development team should conduct a thorough review of the stage-1 classifier's performance and ensure that it is effectively filtering out irrelevant complaints.
2. The data science team should address the limitation of weak supervision through human-adjudicated labels or more robust labeling approaches.
3. The model development team should continuously monitor the stage-1 classifier's performance to ensure that it remains fair and effective.
4. The data science team should address the low recall for individual labels through more robust labeling approaches or human-adjudicated labels.
5. The model development team should establish a regular monitoring schedule to ensure that the model's performance is continuously evaluated.
6. The data science team should address the limitation of weak supervision through human-adjudicated labels or more robust labeling approaches to ensure that the model's complaint routing is fair and effective.

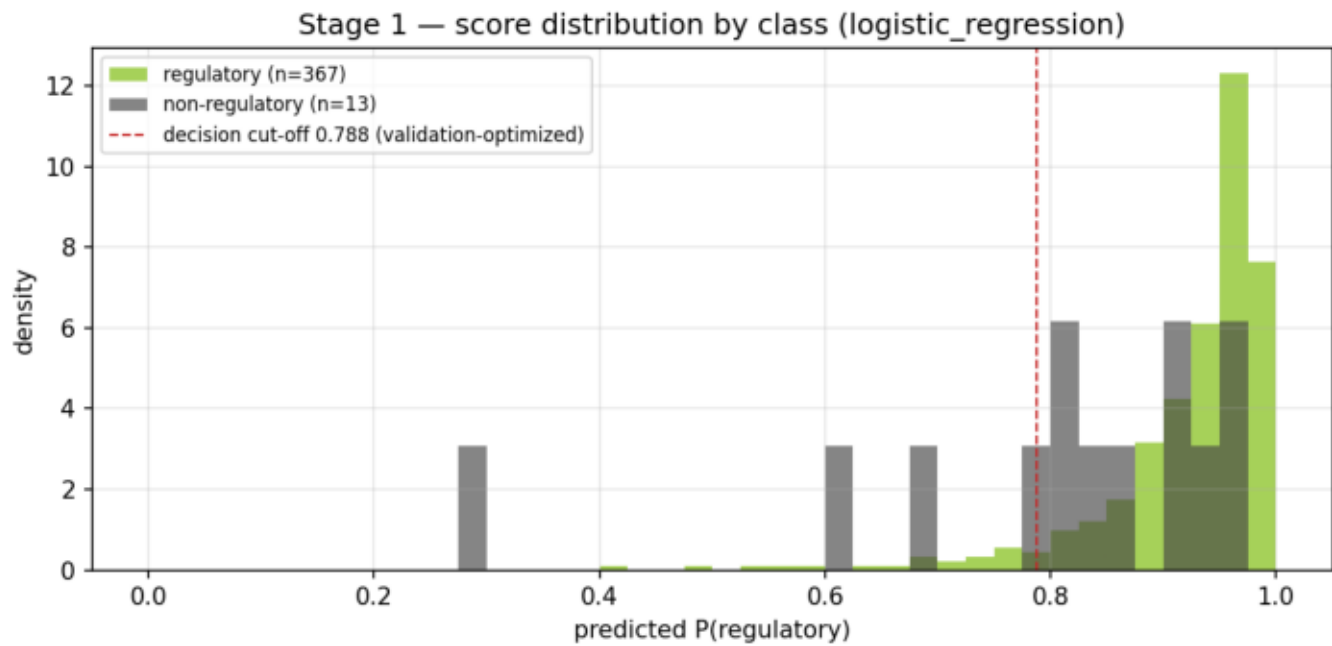
2 · Stage-1 leaderboard (validated re-run)

model	val_pr_auc	threshold	pr_auc	roc_auc	f1	precision	recall	accuracy
logistic_regress	0.9913	0.788	0.9905	0.7965	0.9542	0.9718	0.9373	0.9132
xgboost	0.9901	0.844	0.9882	0.7355	0.9453	0.974	0.9183	0.8974

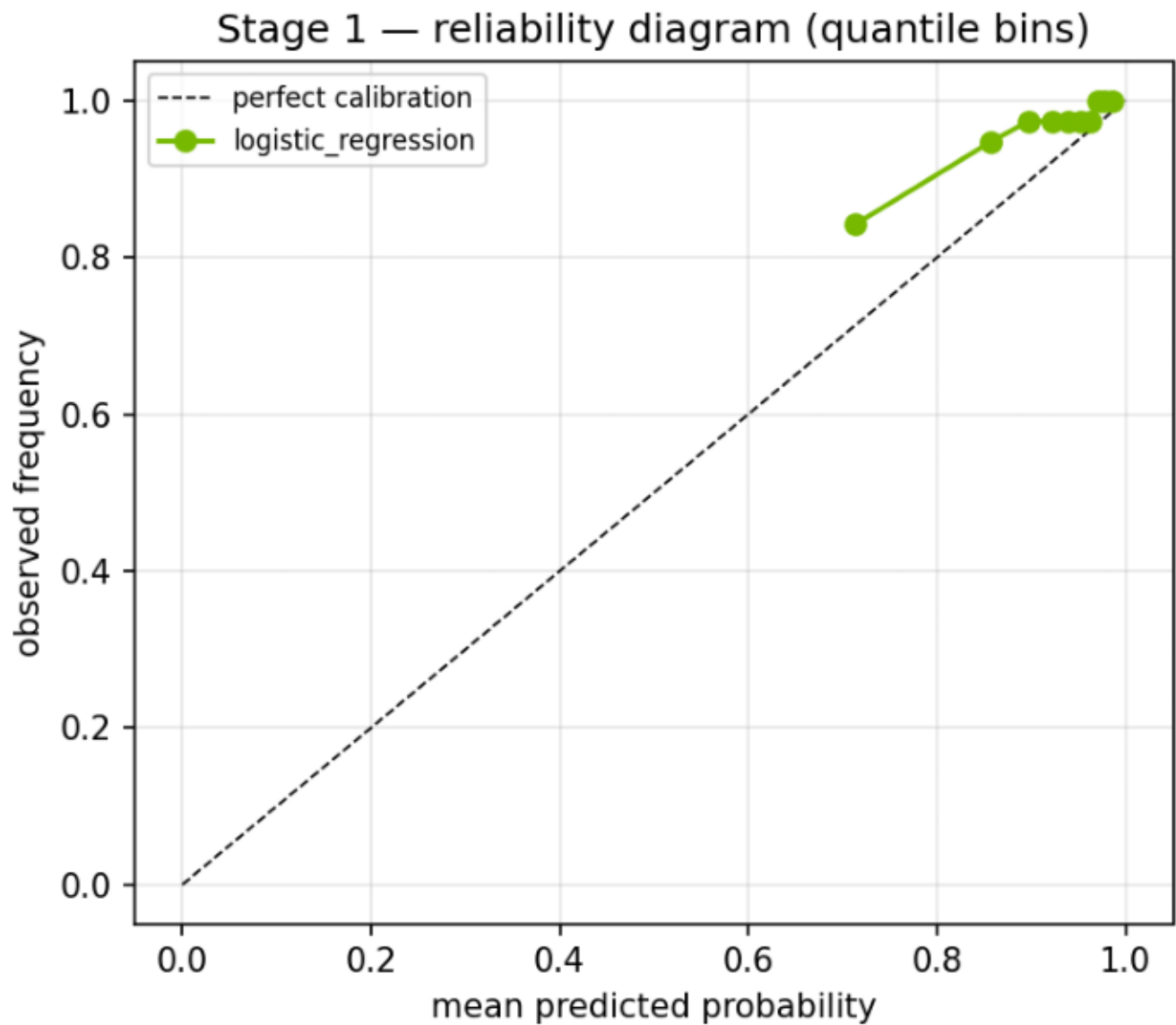
2 · Stage-1 evidence: discrimination



2 · Stage-1 evidence: score separation

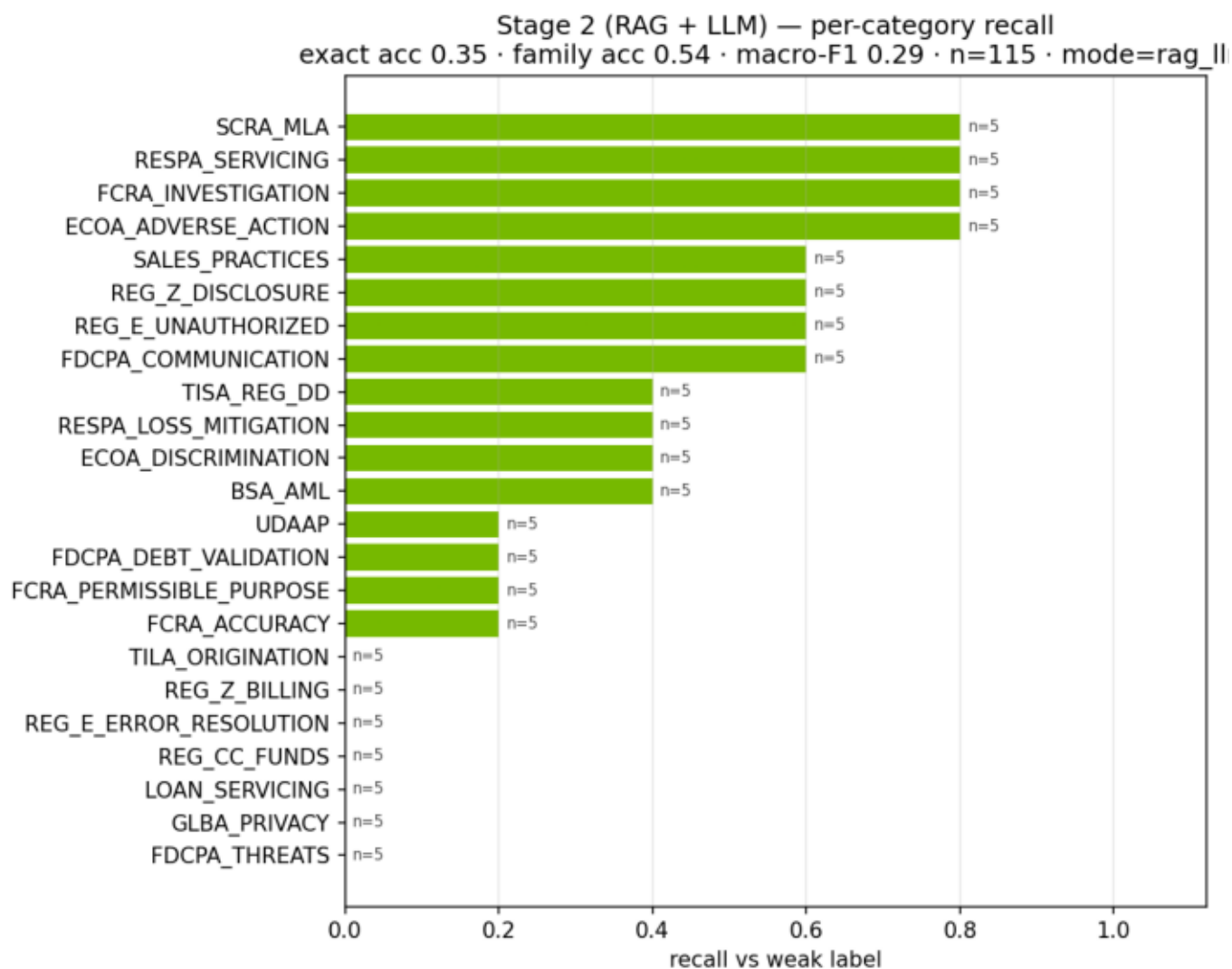


2 · Stage-1 evidence: calibration



Validator re-derived reliability diagram (quantile bins).

3 · Stage-2 evidence



3 • Stage-2 per-category detail

category	support	recall
BSA_AML	5	0.4
ECOA_ADVERSE_ACTION	5	0.8
ECOA_DISCRIMINATION	5	0.4
FCRA_ACCURACY	5	0.2
FCRA_INVESTIGATION	5	0.8
FCRA_PERMISSIBLE_PURPOSE	5	0.2
FDCPA_COMMUNICATION	5	0.6
FDCPA_DEBT_VALIDATION	5	0.2
FDCPA_THREATS	5	0.0
GLBA_PRIVACY	5	0.0
LOAN_SERVICING	5	0.0
REG_CC_FUNDS	5	0.0
REG_E_ERROR_RESOLUTION	5	0.0
REG_E_UNAUTHORIZED	5	0.6
REG_Z_BILLING	5	0.0
REG_Z_DISCLOSURE	5	0.6
RESPA_LOSS_MITIGATION	5	0.4
RESPA_SERVICING	5	0.8
SALES_PRACTICES	5	0.6
SCRA_MLA	5	0.8
TILA_ORIGINATION	5	0.0
TISA_REG_DD	5	0.4
UDAAP	5	0.2

3.1 · Guardrail inventory (validated)

guardrail	mechanism	surfaced as
Taxonomy whitelist	LLM label must be one of the 24 codes; else re	complaint_classifications_total{label}
Strict-JSON parsing	malformed LLM output -> deterministic keyword	mode=fallback counter (alertable spike)
Stage-1 gating	non-regulatory volume never reaches the l	cost + attack-surface control
Retrieval grounding	answer must cite a retrieved corpus exce	citation attached to every prediction
Provider portability	OpenA <-> NIM via config; keyword mode if b	mode label per classification

4 · Limitations & disposition

1. Weak labels: ground truth derives from the CFPB issue taxonomy plus keyword rules, not human adjudication. Condition: build a human-adjudicated golden set (≥ 25 per category).
2. Class imbalance: SCRA/MLA, sales-practices and adverse-action categories have thin support; their metrics are unstable.
3. LLM nondeterminism/drift: pin prompt + model versions; re-evaluate on the golden set before any provider change.
4. Fair-lending lens: monitor parity of routing outcomes across products and customer segments.

DISPOSITION: Approve with Conditions. Re-validation triggered by model/prompt change or stage-1 PSI > 0.25 .

Sign-off

Prepared and reviewed under the bank's model risk management framework (SR 11-7 / OCC 2011-12).

Author, development document: Senior Model Developer & Quantitative Lead — PhD (Econometrics), 20 years in model development across credit, capital planning, fraud, and NLP.

Author, validation report: Senior Independent Model Validator — PhD (Econometrics), 20 years in second-line validation; independent of the development team per SR 11-7 organizational-independence requirements.